Yichao Jin, Ph.D.

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Specialization: Behavioral Decision Science / Health Economics / Discrete Choice Experiments / Clinical Trial Methodology

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Education

University of Texas at Dallas May 2026
Ph.D. in Public Policy and Political Economy

- Dissertation: "Intertemporal Choice in Vaccination Timing: Evidence from a Discrete Choice Experiment in Wuhan"
- Committee: Dohyeong Kim (Chair), Richard Scotch, Soyoung Kwon, Khai Chiong.
- Distinction: Dean of Graduate Education Dissertation Research Award (2025).

University of Texas at Dallas May 2026
M.S. in Social Data Analytics and Research

University of Queensland 2019
Master of Development Economics

University of California, Riverside 2017
B.A. in Economics

Research Agenda

My research lies at the intersection of behavioral decision science, health economics, and computational social science. I develop and empirically validate structural models of individual decision-making under uncertainty — using discrete choice experiments (DCEs), intertemporal choice frameworks, and causally-constrained LLM-based synthetic agents — to understand how people navigate trade-offs involving time, risk, and institutional trust. My work has direct applications in vaccine policy, preference-sensitive health-care, and the design of behavioral interventions that improve population health outcomes.

Working Papers

1. **"Empirical-Frontier Regularization for LLM Synthetic Agents: A Preference-Anchored Framework"**
   Working Paper (Target: Value in Health) — **SSRN Link**. Introduces a DCE-grounded empirical-frontier regularization framework for LLM synthetic agents. Using a convex anchoring rule ($\pi = \lambda\pi_{\text{LLM}} + (1-\lambda)P_{\text{static}}$), the Static-BDT Anchor reduces waiting-time monotonicity violations from 62.5% to 4.2% and improves Spearman rank correlation from -0.024 to 0.952. Presented at AIE Summer Conference 2026 and DAISY 2026 (ACM CHIL 2026).

2. **"Value Contamination in Policy Simulation: Measuring How LLM Policy-Value Orientations Shape Predicted Administrative Burden Effects"**
   Target: Health Services Research — **SSRN Link**. Develops an audit framework for measuring value sensitivity in LLM-generated policy simulations. Using administrative burden in vaccination

access as a testbed, evaluates nine LLMs across six model families through policy-value orientation batteries, narrative frame-decomposition, and within-model frame-manipulation experiments.

3. **"Waiting Time as a Behavioral Barrier to Vaccination Uptake: Nonlinear Heterogeneity by Institutional Trust"**
   Under Review at Economic Analysis and Policy — **Link to Full Draft**
   - Developed a structural Discrete Choice Experiment (DCE) framework to analyze the interaction between temporal barriers and institutional trust in vaccination behavior.
   - Identified nonlinear heterogeneity in waiting time sensitivity, providing a behavioral explanation for vaccine hesitancy.

4. **"Cost-Effectiveness of Intismeran Autogene (mRNA-4157/V940) Plus Pembrolizumab Versus Pembrolizumab Alone as Adjuvant Therapy for Resected High-Risk Melanoma: An Early Economic Evaluation Based on KEYNOTE-942"**
   Working Paper (Target: PharmacoEconomics) — **GitHub: cea-intismeran**. Developed a 4-state partitioned survival model (RF/LR/DM/Death) calibrated to 5-year KEYNOTE-942 data. At estimated pricing ($200,000), intismeran plus pembrolizumab is cost-effective with ICER $∼51,600/QALY and P(CE) 94.9% at $100,000/QALY threshold. All 2,000 PSA draws structurally valid (0 ICER-based exclusions).

Research Experience & Affiliations

Doctoral Researcher, Spatial Health AI Research Partnership (SHARP), UT Dallas

- Leading interdisciplinary research on the intersection of artificial intelligence, geospatial analytics, and health policy.
- Managing large-scale datasets and developing computational simulations (R, Python) to evaluate pandemic preparedness.
- Leading development of the Behavioral Digital Twin (BDT) framework, integrating DCE data with causally-constrained LLM agents for synthetic policy simulation.

Graduate Researcher, Behavioral Health Economics Laboratory, UT Dallas

- Conducted high-salience behavioral experiments (N=1,027) analyzing risk-weighted health decisions and institutional trust.

Teaching Experience

Teaching Assistant, School of Economic, Political and Policy Sciences, UT Dallas

- **Quantitative Methods for Policy Analysis (Graduate)**: Led Econometrics labs; specialized in R programming, regression analysis, and causal inference.
- **Health Economics & Public Policy (Undergraduate)**: Developed case studies on health market failures and administrative challenges.
- **Public Policy Analysis (Undergraduate)**: Mentored students in drafting evidence-based policy memos.

Conference Presentations

- **Invited Talk, Duke University PrefER Confab** (August 2026, Durham, NC): Empirical-Frontier Regularization for LLM Synthetic Agents in Vaccination Preference Research.

- **SJDM 2026 Annual Conference (Society for Judgment and Decision Making)**: Value Contamination in Policy Simulation: Measuring How LLM Policy-Value Orientations Shape Predicted Administrative Burden Effects.
- **AI and Economics (AIE) Summer Conference 2026**: Empirical-Frontier Regularization for LLM Synthetic Agents.
- **APPAM Fall Research Conference 2025** (Seattle, WA): Estimating Time Discount Rate for Vaccination.
- **DAISY 2026 Workshop at ACM Conference on Health, Inference, and Learning (CHIL) 2026**: Behavioral Digital Twins — A Causally-Faithful Generative Framework.
- **UT Dallas Graduate Research Symposium 2025**: Modeling Vaccination Decisions with Hyperbolic Discounting.

Honors & Awards

- **2025** Dean of Graduate Education Dissertation Research Award, UT Dallas
- **2025** Betty & Gifford Johnson Graduate Travel Award, UT Dallas
- **2016** Omicron Delta Epsilon, International Honor Society for Economics

Technical Skills & Certifications

Econometrics: Mixed Logit (MXL), Latent Class Models (LCM), SEIR Modeling, Causal Inference.
AI & ML: PyTorch, Transformers (HuggingFace), LLM APIs (OpenAI, Anthropic), LangChain, Causal Machine Learning.
Software: Stata (Advanced), R, Python, MATLAB, LaTeX, SQL, GIS.
Certifications: Stanford Artificial Intelligence Graduate Certificate; CITI Human Subjects Protection.

REFERENCES

Dohyeong Kim (Chair)
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Robert E. Holmes Jr. Professor of Public Policy, GIS & Social Data Analytics
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